Uncertainty principles and complex analysis

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1. Motivation: Uncertainty à la Heisenberg

In this lecture we will give an overview of the foundations of "uncertainty principles", giving a heuristic in the operator case on Hilbert spaces used in quantum mechanics and focusing on the problem in the case of the Fourier transform. The proofs of Hardy's uncertainty principle and of the Paley-Wiener theorem will allow us to see at work theorems that extend the maximum modulus principle of complex analysis.

1.1 Ideas from quantum mechanics and uncertainty

One of the major results of quantum mechanics is to provide a mathematical framework for principles that cannot be explained in traditional classical mechanics. One of the best known is the *Heisenberg uncertainty principle*, which imposes a lower bound on the product of pairs of variances of observables. The first step is to define how a system, its states and its observables are represented in a quantum theory.

Postulate 1.1: Physical states and observables in a quantum system

A quantum system is associated with a Hilbert space $\mathcal{H}$, whose elements $|\psi\rangle \in \mathcal{H}$ represent the different states in which the system can be found. To every observable is instead associated an operator $\hat{\mathcal{A}} : D \to \mathcal{H}$ that is linear, densely defined and self-adjoint. Moreover:

$$\langle \hat{\mathcal{A}} \rangle_\psi = \frac{\langle \psi | \, \hat{\mathcal{A}} \, | \psi \rangle}{\langle \psi \mid \psi \rangle}.$$

Here a Hilbert space is a vector space over the complex field endowed with a scalar product that makes it complete (in the topological sense), written as $\langle \phi \mid \psi \rangle$. The classic example to keep in mind for this association is the case of the free particle (with spin 0). In this case $\mathcal{H} = L^2(\mathbb{R}^3)$ is the space of position wave functions and $\langle \phi \mid \psi \rangle = \int \bar{\phi}(x)\psi(x)\mathrm{d}^3x$. As for the definition of self-adjointness, speaking somewhat loosely it means that the relation $\langle \phi \mid \hat{\mathcal{A}}\psi \rangle = \langle \hat{\mathcal{A}}\phi \mid \psi \rangle \quad \forall \phi, \psi$[1] holds. At this point we have the means to state the uncertainty principle (theorem).

Theorem 1.1: Heisenberg uncertainty, operator form

Let A and B be two self-adjoint operators; for every $a, b \in \mathbb{R}$ and $|\psi\rangle \neq 0$:

$$(\Delta A)_{a,\psi}(\Delta B)_{b,\psi} \geq \frac{1}{2}|\langle [A, B] \rangle_\psi|. \tag{1.1}$$

Where $(\Delta A)_{a,\psi} = \frac{\langle \psi | (A-a)^2 | \psi \rangle}{\langle \psi | \psi \rangle}$.

The emblematic example of this relation is obtained by considering, in the single-particle example, $A = X \mid \; X\psi(x) = x\psi(x)$ and $B = P \mid \; P\psi(x) = -i\hbar\partial_x\psi(x)$. From the algebraic computation one easily checks that $|\langle [A, B] \rangle_\psi| = i\hbar$, and that consequently one can write:

$$(\Delta X)_{a,\psi}(\Delta P)_{b,\psi} \geq \frac{1}{2}\hbar, \quad \forall a, b, \psi.$$

[1] To this equality one should add conditions on the domain of the adjoint, which we neglect here.

In words, what is written above says that one cannot fix with arbitrary precision, at the same time, the uncertainty on the momentum and on the position of a particle.

1.2 Fourier transform and uncertainty

Let us try to obtain an inequality of the type (1.1) in a simpler setting than that of Hilbert spaces and operators. Consider functions $f : \mathbb{R} \to \mathbb{C}$ and define the Fourier transform $\hat{f} : \mathbb{R} \to \mathbb{C}$ as:

$$\hat{f}(\xi) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i \xi x} \mathrm{d}x.$$

The Fourier transform is a very powerful tool, used in every field of mathematics. What the transform does is to define a weight associated with the original function, which quantifies how much the latter is made of the wave $e^{-2\pi i \xi x}$. Indeed, under suitable assumptions[2], it is possible to rewrite f in terms of its transform, writing:

$$f(x) = \int_{-\infty}^{+\infty} \hat{f}(\xi) e^{2\pi i \xi x} \mathrm{d}\xi.$$

Having defined the transform, we can rewrite an uncertainty principle that relates the variance of a function to that of its transform:

Theorem 1.2: Heisenberg uncertainty principle, Fourier form

Let f and $\hat{f}$ be functions in $L^2(\mathbb{R})$ with norm 1; then:

$$\left(\int_{\mathbb{R}} |x|^2 |f(x)|^2 \, dx \right) \cdot \left(\int_{\mathbb{R}} |\xi|^2 |\hat{f}(\xi)|^2 \, d\xi \right) \geq \frac{1}{(4\pi)^2}. \tag{1.2}$$

Proof: First of all we observe that the following identity holds:

$$\langle xf, f' \rangle = \int_{\mathbb{R}} x \bar{f}(x) f'(x) \mathrm{d}x = -\frac{1}{2} \int_{\mathbb{R}} |f(x)|^2 \mathrm{d}x = -\frac{1}{2}.$$

On the other hand, applying Cauchy-Schwarz:

$$|\langle xf, f' \rangle|^2 \leq \left(\int_{\mathbb{R}} |xf(x)|^2 \mathrm{d}x \right) \left(\int_{\mathbb{R}} |f'(x)|^2 \, \mathrm{d}x \right).$$

Finally, using Plancherel's identity:

$$\int_{\mathbb{R}} |f'(x)|^2 \, \mathrm{d}x = 4\pi^2 \int_{\mathbb{R}} |\xi f(\xi)|^2 \mathrm{d}\xi.$$

And combining the three equations one obtains the claim. □

[2] A sufficient assumption is that both f and $\hat{f}$ be of moderate decrease, i.e. such that $|f(x)| \leq \frac{A}{1+x^2}$.

2. An extension theorem: Lindelöf's principle

Before going on with this excursus on uncertainty theorems, we show a result that extends the maximum modulus theorem for holomorphic functions in complex analysis, which will be useful later. Let us first recall the maximum modulus theorem:

Theorem 2.1: Maximum modulus on a bounded domain

Let $f(z)$ be an analytic function defined on a bounded domain D and continuous on the boundary ∂D. Then the maximum value of $|f(z)|$ on the closure of D is attained on ∂D.

Proof: Omitted. □

An important limitation of this theorem is that it holds only for bounded sets, while it is not rare to work with unbounded sets. Fortunately, in certain contexts it is possible to relax this assumption, at the price of imposing constraints on the shape of the set and of giving bounds on the growth of the function. The generalisation we will use is due to Phragmen and Lindelöf:

Theorem 2.2: Phragmen and Lindelöf

Let $f(z)$ be a holomorphic function defined on the sector:

$$D = \left\{ z : -\frac{\pi}{4} \lneq arg(z) \lneq \frac{\pi}{4} \right\}.$$

And such that on the boundary ∂D it is continuous and bounded, $|f(z)| \leq 1$. If there exist constants C,c such that: $|f(z)| \leq Ce^{c|z|}, \quad \forall z \in D$. Then:

$$|f(z)| \leq 1 \quad \forall z \in D.$$

We observe that the function $f(z) = e^{z^2}$ satisfies the boundedness assumptions on the boundary, but on the real axis it diverges at infinity. Consequently it is necessary to impose a bound on the overall growth of the function. In fact, as can be seen from the proof, this is the sharp counterexample to the theorem.

Proof: Consider the functions $f_\epsilon(z) = f(z)e^{-\epsilon z^{3/2}}$. Let us first show that this term makes the function $f_\epsilon(z)$ decay, by observing that the modulus of the additional factor is:

$$\left| e^{-\epsilon z^{3/2}} \right| = e^{-r^{3/2} cos(3\theta/2)}.$$

And that the cosine term takes positive values for every $z \in D$, since:

$$-\frac{\pi}{2} \lneq -\frac{3\pi}{8} \leq \frac{3\theta}{2} \leq \frac{3\pi}{8} \lneq \frac{\pi}{2}.$$

Since $f(z)$ is controlled by an exponential growth of power 1, overall the function $f_\epsilon(z)$ decays exponentially in modulus as $z \to \infty$. For this reason the function cannot have its maximum value at infinity, but it must lie in a ball of radius M. Having now a bounded set, we can say from the classical maximum modulus theorem that the maximum will be on the boundary, which is bounded by 1 by assumption (the additional term can at most

reduce the value of this maximum, while the third boundary is negligible provided we take a ball of sufficiently large radius). In this way we have $|f_\epsilon(z)| \leq 1 \quad \forall z \in D, \forall \epsilon > 0$. Taking the limit $\epsilon \to 0$ one obtains the claim. □

The proof of the theorem is easy to generalise, by changing at the same time the size of the sector and the power of the exponential that defines the growth of the function.

Theorem 2.3: Phragmen and Lindelöf 2

Let $f(z)$ be a holomorphic function defined on the sector of angle $\frac{\pi}{\beta}$, and such that on the boundary ∂D it is continuous and bounded, $|f(z)| \leq 1$. If there exist constants C,c such that: $|f(z)| \leq Ce^{c|z|^\alpha}, \quad \forall z \in D$, with $\alpha < \beta$, then:

$$|f(z)| \leq 1 \quad \forall z \in D.$$

Proof: omitted (exercise). □

Finally, we observe that even though the theorem holds for "wedges" of the complex plane, it can also be applied to other sets (among them $\mathbb{C}$ itself), by decomposing them into wedges and checking the theorem on each of its parts.

3. Applications

Let us now see two examples of application of Lindelöf's theorem, namely the Paley-Wiener theorem and Hardy's uncertainty principle. The first one informally says that if one has a weak decay control of a function f on the real axis, while in the rest of the plane the growth is at most exponential, the transform of f must have compact support (i.e. a very fast decay). On the other hand, Hardy's theorem says that $f(x)$ and its transform cannot both decay too fast (where by too fast we mean with Gaussian decay). In particular, the only functions having this double decay property are the Gaussians.
An interesting observation that emerges from the proofs is that the real bounds of the two statements force the growth of $f(z)$ on the complex plane to be indeed exponential, as Lindelöf requires, but with respect to only one of the real variables that make up $z = x + iy$. This fact, almost accidental in the first theorem, will instead prove crucial in the second, where one has to apply the maximum modulus principle precisely in the sharp negative case ($\alpha = \beta$), in which, in the general formulation of the problem, it would turn out to be false.

3.1 Paley Wiener theorem

Theorem 3.1: Paley-Wiener

Let f be a continuous function of moderate decrease on $\mathbb{R}$. Then f has an entire extension to the complex plane with growth bounded by $|f(z)| \leq Ae^{2\pi M|z|}$ if and only if the Fourier transform $\hat{f}$ has support in $[-M, M]$.

Proof: Let us begin by considering the case in which the Fourier transform has compact support. In that case, by the inversion formula one can write :

$$|f(z)| = \left| \int_{\mathbb{R}} e^{2\pi i \xi z} \hat{f}(\xi) \mathrm{d}\xi \right| \leq \int_{-M}^{M} e^{-2\pi \xi y} \left| \hat{f}(\xi) \right| \mathrm{d}\xi \leq Ae^{2\pi M|y|}.$$

We observe that we obtain a better inequality than the one we hoped for, since we do not have to impose any bound on the growth with respect to x. To prove the converse, since we have to show the equivalence of the two conditions, we would actually like to reduce to this weaker setting. To do so we proceed in steps:
Step 1: suppose that f is holomorphic on the complex plane and bounded by:

$$|f(x+iy)| \leq A' \frac{e^{2\pi M|y|}}{1+x^2}.$$

Writing the Fourier transform:

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2\pi i x \xi} f(x) \mathrm{d}x = \int_{\mathbb{R}} e^{-2\pi i (x-iy)\xi} f(x-iy) \mathrm{d}x$$
$$\left| \hat{f}(\xi) \right| \leq \int_{\mathbb{R}} \frac{A' e^{-2\pi |y|(\xi - M)}}{1+x^2} \mathrm{d}x \leq A'' e^{-2\pi |y|(\xi - M)}$$

Since what is written holds $\forall y$, it is clear that if $\xi - M > 0$ we obtain $\hat{f}(\xi) = 0$.
Step 2: Now assume that f satisfies:

$$|f(x+iy)| \leq Ae^{2\pi M|y|}.$$

Consider now the sequence of functions indexed by ϵ:

$$f_\epsilon(z) = \frac{f(z)}{(1+i\epsilon z)^2}$$

Such that $f_\epsilon \to f$ in the limit $\epsilon \to 0$. In particular, by the moderate-decrease assumption on f one easily sees that the transforms $\hat{f}_\epsilon$ also converge to $\hat{f}$. But at this point it is enough to observe that in the complex half-plane with $y < 0$ the f_ϵ are functions of the type assumed in step 1 (at fixed epsilon), and therefore f is zero, being the limit of a zero sequence (the opposite half-plane is obtained by taking $(1 - i\epsilon z)^2$).

$$\frac{1}{|1+i\epsilon z|^2} \leq \frac{1}{|1+i\epsilon z|^2} \leq \frac{1}{(1-\epsilon y)^2 + (\epsilon x)^2} \leq \frac{1}{\epsilon^2} \frac{1}{1+x^2}.$$

Step 3: Let us now place ourselves under the hypotheses of the theorem and try to show that, given the bound on the growth of the function and the hypothesis $|f(x)| \leq 1$ (from moderate decrease), we can force the inequality $|f(z)| \leq Ae^{2\pi M|y|}$. Consider the new function $F(z) = f(z)e^{2\pi i M z}$. At this point I have:

- $|F(x)| \leq 1$ by moderate decrease;
- $|F(y)| \leq 1$ if $y > 0$ because of the additional term of F;
- in the first quadrant I have decay $e^{2\pi M|z|}$.

That is, in the first quadrant $F(z)$ satisfies the hypotheses of Lindelöf's theorem, so that:

$$|F(z)| \leq 1 \to |f(z)| \leq e^{2\pi i M|y|}.$$

In the whole first quadrant. Considering that the other quadrants can be treated in an analogous way (by changing the definition of $F(z)$), we are back under the hypotheses of step 2, and this concludes the proof. □

3.2 Uncertainty à la Hardy

Theorem 3.2: Hardy's uncertainty principle

Let $f(x)$ be a measurable function such that $|f(x)| \leq Ce^{-\pi a x^2}$ and $\left|\hat{f}(\xi)\right| \leq C' e^{-\pi \xi^2/a}$ for some $C, C' > 0$. Then $f(x)$ is a multiple of the Gaussian $e^{-\pi a x^2}$.

Proof: By dilations we can assume WLOG $a = C = 1$. Let us now use the Gaussian decay property to extend the Fourier transform to the complex plane. Indeed one obtains the estimate:

$$\left|\hat{f}(\xi + i\eta)\right| = \left|\int_{\mathbb{R}} e^{-2\pi i x(\xi+i\eta)} f(x) \mathrm{d}x\right| \leq \int_{\mathbb{R}} e^{2\pi x\eta} |f(x)| \mathrm{d}x \leq e^{\pi\eta^2}.$$

Consider now the new function $F(z) = \hat{f}(z)e^{\pi z^2}$ and observe that:

- By hypothesis, on the real axis $F(x) \leq 1$;
- On the imaginary axis, by what we found above, $F(iy) \leq 1$;
- in the first quadrant I have a Gaussian growth $F(\xi + i\eta) \leq e^{\pi\xi^2}$.

These are almost the hypotheses needed to apply Lindelöf's theorem. If the theorem were applicable, we would have that $F(z)$ is entire and bounded by 1, hence by Liouville's theorem it is constant. But then $\hat{f}(z) = Ae^{-\pi z^2}$, from which the claim follows. In reality we are in the limiting case in which Lindelöf's theorem does not apply, since we consider a quadrant of angle $\pi/2$ and a growth exponent 2 (whereas <2 would be needed). On the other hand, thanks to the fact that the decay of the transform is uniform in x, we can overcome this problem. Consider the problem in the sector of angle $0 < \alpha < \theta < \pi/2$ and define a new function $\tilde{F}(z) = F(z)e^{i\delta z^2}$. This new function, provided the angle is sufficiently close to 90°, has its boundary bounded by 1 (at the angle θ the additional term produces a decay of $e^{-2\delta\eta\xi}$, which counterbalances the factor $e^{\pi\xi^2}$) and its interior controlled with the same estimates as $F(z)$. Having restricted the angle, the growth is sufficient to apply Lindelöf's theorem and say that the function in the sector is bounded by 1. Letting $\theta \to \pi/2$, $\delta \to 0$ and repeating the same argument for all four quadrants one obtains the claim. □

Exercise: show that if $|f(x)| \leq Ae^{-\pi ax^2}$ and $\left|\hat{f}(\xi)\right| \leq A'e^{-\pi b\xi^2}$ with $ab > 1$, then $f(x)$ is identically zero.

An interesting observation is that, in its formulation, Hardy's theorem makes no use of complex analysis, which turns out to be a tool for approaching the problem. Nevertheless, proofs of the problem that use real analysis only (which becomes necessary when the transform is generalised to $\mathbb{R}^n$) turn out to be much more laborious. Finally, we observe that this theorem is called an uncertainty principle because, as described in the previous examples, it is a statement showing that a function and its transform cannot both be very concentrated in the domain at the same time, just as the variances showed in the operator case of quantum mechanics.

4. References

- For uncertainty à la Heisenberg (Fourier) and à la Hardy: "https://terrytao.wordpress.com /2009/02/18/hardys-uncertainty-principle/";
- For Paley Wiener and Lindelöf's principle: Complex Analysis, Stein and Shakarchi;
- Introduction to quantum mechanics: Mastering quantum mechanics, Zwiebach.